Random Forests

An Introduction with Regression and Classification Applications

Ryan McMurray

Melanie Jensen

Lucca Coelho

Math 3190: Fundamentals of Data Science

Rick Brown

Southern Utah University

April 22, 2026

Mathematical Foundation and Assumptions

Random forests are an ensemble learning method that constructs a large collection of decision trees and aggregate their predictions to produce a single model. They are widely used for both regression and classification problems, particularly in settings where the relationship between predictors and the response is nonlinear or involves complex interactions. The central motivation for random forests is the reduction of variance: while individual decision trees tend to be highly variable, averaging many such trees, provided they are sufficiently decorrelated, yields a more stable and accurate predictor.

The construction of a random forest begins with decision trees, which partition the feature space into a set of disjoint regions and assign predictions based on those regions. In the regression setting, a tree partitions the predictor space into regions $R_1, \ldots, R_J$ and defines a piecewise constant function of the form $\hat{f}(x) = \sum_{j=1}^{J} c_j \mathbf{1}(x \in R_j)$, where c_j is typically the mean of the observed responses within region R_j . These regions are chosen by recursively splitting the data to minimize the residual sum of squares. In the classification setting, predictions are instead made by majority vote within each region, and splits are chosen to reduce impurity, commonly measured by the Gini index or entropy. These criteria quantify how mixed the class labels are within a node, encouraging splits that produce more homogeneous subsets.

A random forest extends this framework by introducing randomness into the tree-building process. Specifically, each tree is trained on a bootstrap sample of the data, meaning that observations are sampled with replacement to form a new dataset of the same size. In addition, at each split within a tree, only a random subset of the available predictors is considered. This second source of randomness is critical, as it reduces the correlation between individual trees. Without it, trees built on bootstrap samples alone tend to be similar, particularly when there are strong predictors, limiting the effectiveness of averaging.

Predictions from a random forest are obtained by aggregating the outputs of the individual trees. For regression, this is done by averaging, $\hat{f}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$, while for classification, the predicted class is determined by majority vote across the trees. The theoretical benefit of this approach can be understood through the variance of the ensemble. If each tree has variance σ^2 and pairwise correlation ρ , then the variance of the averaged predictor is approximately $\mathrm{Var}(\hat{f}) \approx \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. This expression highlights that increasing the number of trees reduces variance, but the reduction is most effective when the trees are weakly correlated. The random feature selection mechanism plays a key role in achieving this decorrelation.

Although random forests are often described as making few assumptions, they do rely on several important conditions. The training observations are assumed to be independent and identically distributed, and the model implicitly assumes that the distribution of the training data matches that of future observations. Furthermore, the predictors must contain sufficient information about the response for meaningful patterns to be learned. Like other tree-based methods, random forests are also limited in their ability to extrapolate beyond the range of the observed data, as predictions are constructed from averages of observed outcomes.

An additional practical advantage of random forests is the availability of out-of-bag error estimation. Because each tree is trained on a bootstrap sample, approximately one-third of the observations are omitted from the training set for that tree. These unused observations can be used to evaluate predictive performance by averaging predictions over the subset of trees in which the observation was not included. This provides an efficient and nearly unbiased estimate of generalization error without the need for a separate validation set.

Despite their strong empirical performance, random forests are not without limitations. Their ensemble nature makes them less interpretable than simpler models, and commonly used measures of feature importance can be difficult to interpret or biased in the presence of correlated predictors.

Additionally, training a large number of trees can be computationally demanding, particularly for large datasets or high-dimensional problems. Random forests may also struggle when many predictors are irrelevant, and, as noted earlier, they do not extrapolate well beyond the range of the training data.

In summary, random forests build upon the strengths of decision trees while addressing their primary weakness—high variance—through averaging and randomization. By combining bootstrap sampling with random feature selection, they produce a flexible and robust modeling approach that performs well across a wide range of regression and classification problems.

Regression Example

To demonstrate how random forest regression behaves in practice, this section walks through the process of building, tuning, and evaluating a random forest regression model on the Ames Housing dataset. This dataset contains 2,930 observations on residential home sales in Ames, Iowa, along with 80 predictor variables covering physical characteristics, quality ratings, location, and sale conditions. The response variable of interest is `Sale_Price`, the final sale price of each home in dollars. The dataset is well suited to a random forest demonstration because it contains a mix of continuous and categorical predictors, several groups of correlated variables, and nonlinear relationships that would be difficult to capture with a linear model without substantial feature engineering.

Setup and Data Preparation

The analysis uses the `randomForest` package in R for model fitting and tuning, along with `tidyverse` packages for data manipulation and visualization. The `AmesHousing` package provides a cleaned version of the data through its `make_ames()` function, which handles factor level encoding and removes the need for most preprocessing.

```r
library(tidyverse)
library(randomForest)
library(AmesHousing)
library(kableExtra)

ames <- make_ames()

set.seed(2026)
split <- sample(nrow(ames), size = 0.8 * nrow(ames))
train <- ames[split, ]
test <- ames[-split, ]
```

An 80/20 train test split was used, with 80 percent of the data reserved for model fitting and the remaining 20 percent held out for evaluation. A fixed random seed was set before the split to ensure reproducibility, which is especially important for random forests given that both the bootstrap sampling process and the random feature selection at each split introduce stochasticity into the model.

Fitting the Initial Model

The initial random forest model was fit using all 80 predictors to demonstrate one of the practical strengths of the method, which is its ability to handle high-dimensional data without requiring manual variable selection or transformation. The model was built with the default 500 trees and `mtry` set to the regression default of `p/3`, where `p` is the number of predictors. This default corresponds to 26 variables being randomly considered at each split in each tree.

```r
set.seed(2026)
rf_model <- randomForest(
  Sale_Price ~ .,
  data = train,
  ntree = 500,
  mtry = floor((ncol(train) - 1) / 3),
  importance = TRUE
)
```

The `importance = TRUE` argument instructs the function to compute permutation-based variable

importance during training, which will be used in a later section to examine which predictors the model is relying on most. Computing importance during the initial fit avoids the need to refit the model later solely for that purpose.

The output of the fitted model reports that 500 trees were built with 26 variables considered at each split, an out-of-bag mean squared error of approximately 6.26×10^8, and a percent of variance explained of roughly 89 percent. These out-of-bag statistics come from a built-in validation mechanism specific to bagged methods. Because each tree is trained on a bootstrap sample that omits roughly one third of the training observations, those omitted observations can be used to evaluate the prediction quality of each tree without requiring a separate validation set. Aggregating these predictions across all trees in which an observation was out-of-bag produces a nearly unbiased estimate of generalization error, which is reported as the out-of-bag MSE.

Determining the Appropriate Number of Trees

Although 500 trees is the default, the choice of `ntree` is worth examining because it has implications for both model stability and computational cost. Unlike most hyperparameters, `ntree` is not a tuning parameter in the traditional sense. Adding more trees cannot cause a random forest to overfit, because each additional tree simply refines the average prediction rather than increasing model complexity. The question is therefore not which value of `ntree` minimizes error, but rather how many trees are needed for the out-of-bag error to stabilize.

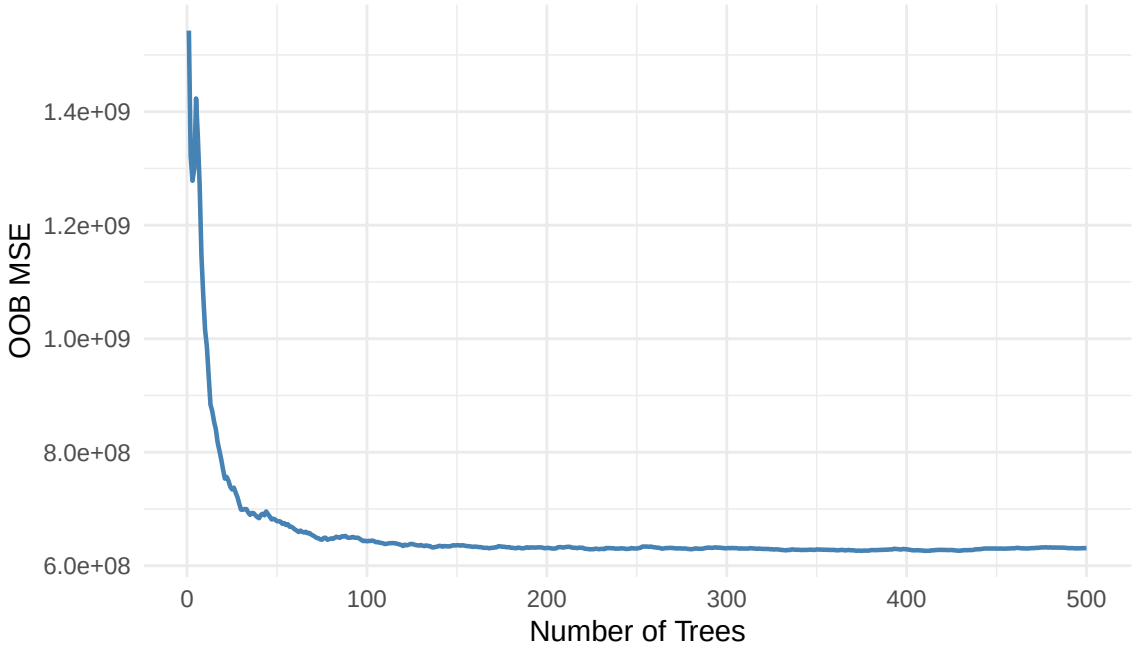


Figure 1: Out-of-bag mean squared error as a function of the number of trees in the forest.

Figure 1 shows how the out-of-bag MSE evolves as trees are added to the forest. The error drops rapidly over the first 100 trees and stabilizes near 200 trees, after which additional trees produce no meaningful reduction in error. This pattern is consistent with the theoretical behavior of bagged ensembles, where variance is reduced by averaging but the benefit diminishes as more trees are added.

Tuning the Number of Variables Per Split

The more meaningful hyperparameter in a random forest is `mtry`, which controls how many predictors are randomly considered as candidates at each split. Lower values produce more decorrelated trees at the cost of individual tree accuracy, while higher values allow each tree to consider stronger splits but make the trees more similar to one another. The `tuneRF` function in the `randomForest` package searches across `mtry` values by adjusting the starting value up and down by a chosen step factor and selecting the value that minimizes out-of-bag error.

```r
set.seed(2026)
tuneRF(
  x = train |> select(-Sale_Price),
  y = train$Sale_Price,
  ntreeTry = 200,
  stepFactor = 1.5,
  improve = 0.01,
  trace = FALSE,
  plot = TRUE
)
```

The tuning procedure tested `mtry` values of 18, 26, and 39. The default value of 26 produced the lowest out-of-bag MSE, with the alternatives performing less than one percent worse. This result illustrates a well-known property of random forests, which is their insensitivity to hyperparameter choices within a reasonable range. The default `p/3` heuristic was confirmed to be near-optimal for this dataset, and no additional tuning was warranted.

A final model was fit using the tuned specification. Because the convergence analysis in the previous section showed that out-of-bag error stabilized well before 500 trees, the final model was built with only 200 trees to reduce computational cost while preserving essentially identical predictive performance.

```r
set.seed(2026)
tuned_rf_model <- randomForest(
  Sale_Price ~ .,
  data = train,
  ntree = 200,
  mtry = 26,
  importance = TRUE
)
```

Model Evaluation

Evaluation of a random forest regression model differs meaningfully from evaluation of a linear regression model. Because random forests make essentially no distributional assumptions about the data or the errors, traditional diagnostic procedures such as checking for normality of residuals or

constant variance do not apply in the same way. Instead, evaluation focuses on direct measures of predictive performance on held-out data and on visual inspection of where the model performs well or poorly.

Table 1: Performance metrics for the tuned random forest on held-out test data.

Metric	Value
Test RMSE	\$25,154
Test R^2	0.911
OOB RMSE	\$25,127

The test RMSE of approximately \$25,154 indicates that on average the model predictions are off by around that amount on houses with a mean sale price of roughly \$180,000, which corresponds to about a 14 percent relative error. The test R^2 of 0.911 indicates that the model explains 91 percent of the variance in sale price on houses it was never trained on. Perhaps most notably, the out-of-bag RMSE of \$25,127 differs from the test RMSE by only \$27, which confirms that the out-of-bag error provides a reliable estimate of generalization performance. This close agreement is a practical validation of the theoretical claim that out-of-bag estimates are nearly unbiased approximations of test error, and it suggests that in settings where holding out a test set is undesirable, the out-of-bag estimate alone would have produced essentially the same conclusion about model quality. This close alignment also serves as a robustness check on the method itself, confirming that the training procedure produced a model that generalizes consistently rather than one that happened to fit the training data well by chance. For random forest regression, this kind of internal validation is a key diagnostic tool, particularly when working with smaller datasets where setting aside a test set would meaningfully reduce the information available for model fitting.

Figure 2: Predicted versus actual sale prices on the held-out test set. The dashed red line represents perfect prediction.

Figure 2 plots predicted sale prices against actual sale prices on the test set. In a well-performing model, points should cluster tightly around the diagonal line with no systematic drift above or below it, which would indicate that predictions are accurate across the entire range of the response. Points cluster tightly around the diagonal through most of the price range, with the densest region of data falling between $100,000 and $300,000 where the model performs best. Two patterns become visible at the upper end of the price range. The spread of the points widens, indicating that prediction uncertainty grows for more expensive homes, and the points shift systematically below the diagonal, indicating that the model tends to underpredict the most expensive homes. Both patterns share a common cause. Random forest predictions are constructed by averaging responses in the terminal nodes of each tree, which means no prediction can exceed the largest value observed in the training data. For homes at or beyond the upper range of the training set, the model is both less informed by data and structurally incapable of extrapolating upward. This is a known limitation of tree-based

methods and is important to acknowledge when applying random forests to problems where extreme values matter.

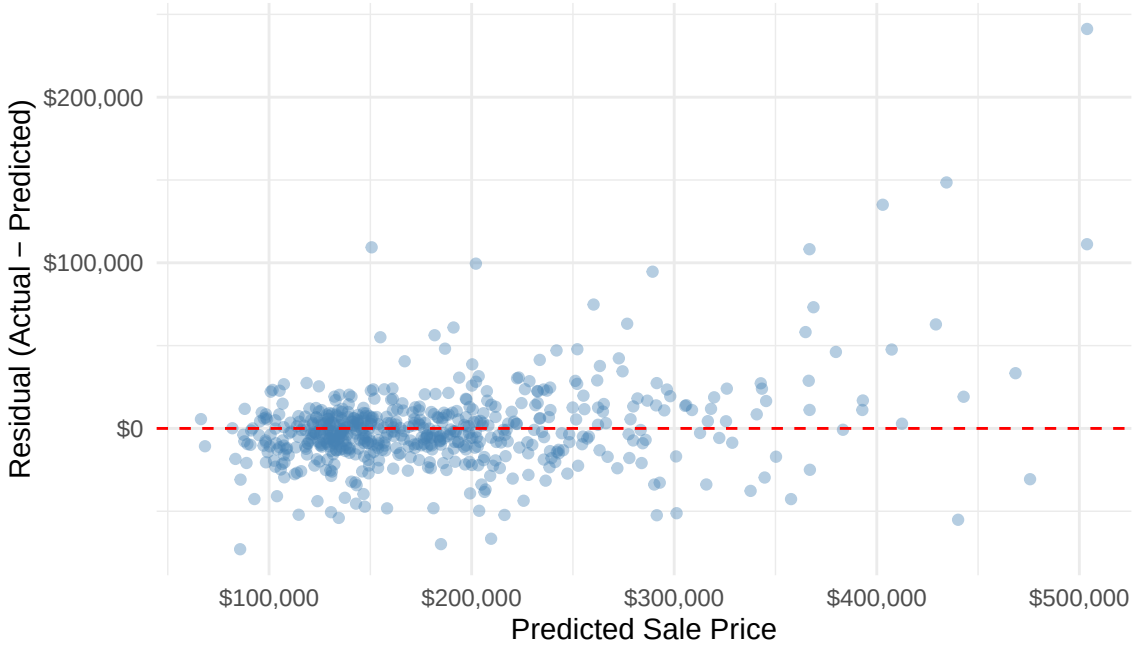


Figure 3: Residuals from the tuned random forest model, defined as actual minus predicted sale price, plotted against predicted sale price.

The residual plot in Figure 3 makes the same patterns explicit. Residuals fan out as predictions increase, and at the highest predicted values the residuals are predominantly positive, which is the mirror image of the underprediction visible in Figure 2. In a linear regression context, this pattern would be interpreted as a violation of the homoscedasticity assumption and would prompt remedial action such as a log transformation of the response. In the random forest context, where no such assumption is being made, the pattern is instead interpreted descriptively as evidence that the model performs less reliably at the extremes of the price distribution.

Variable Importance

One of the practical advantages of random forests over linear models is the availability of built-in variable importance measures that describe which predictors the model is relying on most. This

analysis uses permutation-based importance, reported as percent increase in mean squared error when a variable is permuted. The procedure takes the out-of-bag observations and randomly shuffles the values of a single predictor while holding all other predictors fixed, then measures how much the out-of-bag MSE increases as a result. A variable whose permutation causes a large increase in MSE is one the model is relying on heavily for accurate predictions, while a variable whose permutation changes MSE very little is one the model is effectively ignoring.

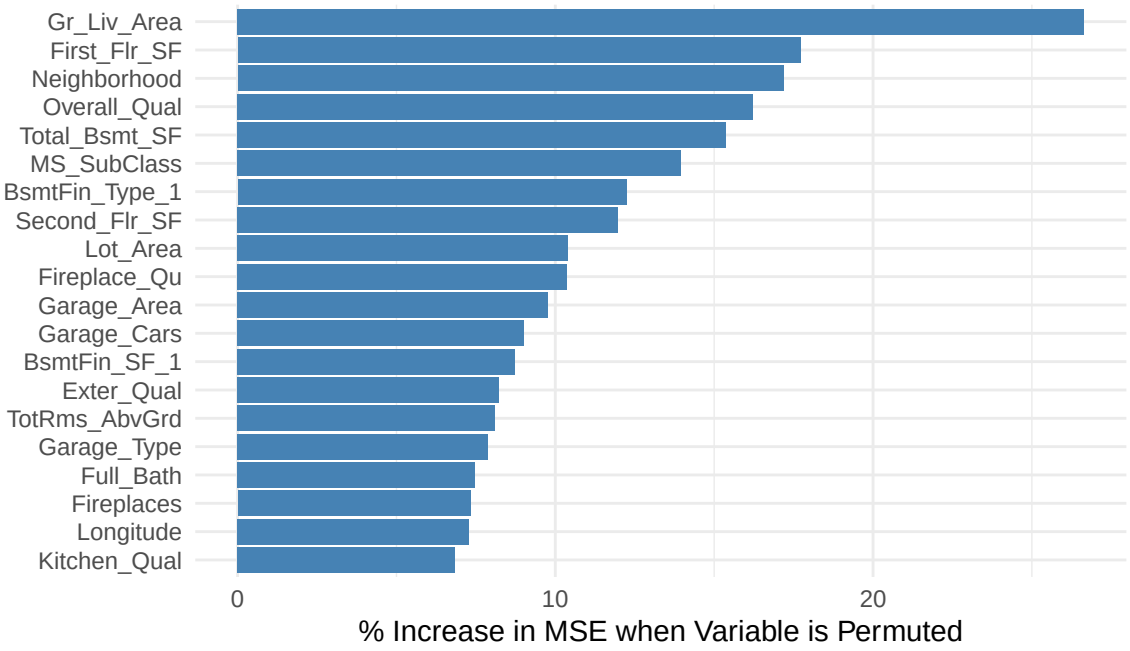


Figure 4: Top 20 predictors ranked by permutation-based variable importance, measured as the percent increase in out-of-bag MSE when the variable is permuted.

Figure 4 shows the top 20 predictors ranked by permutation importance. The most important variables are `Gr_Liv_Area` (above-ground living area), followed by `First_Flr_SF`, `Neighborhood`, `Total_Bsmt_SF`, and several other size and location variables. This ranking aligns with real-world intuition about housing prices, where square footage and location are typically the strongest drivers of value. The alignment between the model's most important variables and what a domain expert would prioritize serves as a useful sanity check that the model is learning meaningful patterns rather

than spurious ones.

Below the top 20 predictors, importance scores drop toward zero, and a number of variables show slightly negative values. A negative percent increase in MSE does not mean that the variable actively harms predictions. Instead, it reflects the fact that the permutation procedure introduces its own random variation, and when a variable contributes essentially no predictive signal, the measured importance fluctuates around zero and can land on either side. In practical terms, variables at the bottom of the ranking are either carrying information already captured by correlated predictors or have no meaningful relationship with sale price, and the model is not relying on them regardless.

This finding has a useful implication for model simplification. Because many of the 80 predictors in the model contribute little to prediction, a reduced model fit on only the top 15 to 20 predictors would likely perform nearly as well as the full model. In settings where interpretability, training speed, or data collection cost matter, using variable importance to guide feature selection is a common and defensible practice. For the purposes of this demonstration, the full model is retained to illustrate how random forests handle high-dimensional data without requiring that selection step.

Summary

The random forest regression model built on the Ames Housing dataset achieves strong predictive performance on held-out test data, explaining approximately 91 percent of the variance in sale price with a test RMSE of about $25,000. The close agreement between out-of-bag and test error confirms the theoretical property that out-of-bag estimation provides a reliable approximation of generalization error. Tuning analysis showed that the default hyperparameters were near-optimal, consistent with the well-known robustness of random forests to parameter choices. Variable importance analysis produced a ranking that aligns with domain intuition, with size and location predictors dominating the model. The primary limitation observed was the model's tendency to underpredict

the highest-priced homes, which reflects the structural inability of tree-based methods to extrapolate beyond the training data. Overall, the example demonstrates that random forests are well suited to problems with many mixed-type predictors, nonlinear relationships, and correlated features, where linear models would require substantial preprocessing to perform comparably. The method is most appropriate in applied settings where predictive accuracy is the primary concern, such as price prediction, risk scoring, ecological modeling, and medical diagnostics. It is particularly valuable when working with datasets that contain a large number of candidate predictors whose individual relationships with the response are not well understood, because the algorithm can identify informative variables without requiring the analyst to specify those relationships in advance. The method is less appropriate when the goal is formal statistical inference or when predictions must extend beyond the range of the observed data, in which case a parametric model with explicit assumptions about the relationship between predictors and the response would be more suitable.

Classification Example

To illustrate how random forest performs in a classification setting, this section applies the algorithm to the very well known iris dataset. The dataset consists of 150 observations of iris flowers, each described by a numerical feature: sepal length, sepal width, petal length, and petal width. The response variable is the species of the flower, which contain three classes: setosa, versicolor, and virginica. This dataset is very useful to demonstrate the classification techniques because the species are linearly separable but also shows some overlap, making it simple enough to visualize yet complex enough to show the strengths of more flexible models.

In contrast to regression, where the goal is to predict a continuous numerical value, classification focuses on assigning observations to categories. Random forests handle this the way you normally expects of this model, by building multiple decision trees that each produce a class prediction, and

then is a matter of majority wins. The iris dataset provides a clear setting to observe how random forests partition the feature space and improve predictive accuracy compared to individual trees, while also allowing for straightforward evaluation using metrics such as accuracy and confusion matrices.

Setup and Data Preparation

```
library(randomForest)
library(caret) |> suppressPackageStartupMessages()
library(ggplot2)
library(dplyr)
library(tidyr)
library(forcats)
library(reprtree)

data(iris)

trainIndex = sample(1:nrow(iris), 0.7 * nrow(iris))

trainData = iris[trainIndex, ]
testData = iris[-trainIndex, ]
```

To evaluate model performance, the data was randomly split into training and testing sets using an 70/30 split, where 70 percent of the observations were used to train the model and the remaining 30 percent were held out for testing, the split was 70/30 instead of the 80/20 used in regression because the iris dataset is significantly smaller. This separation ensures that model performance is assessed on unseen data, providing a more realistic measure of how well the classifier generalizes beyond the training sample.

Building a Model

```
Call:
 randomForest(formula = Species ~ ., data = trainData, type = "classification",      importance
               Type of random forest: classification
                     Number of trees: 500
```

```
No. of variables tried at each split: 2

        OOB estimate of  error rate: 1.9%
Confusion matrix:
           setosa versicolor virginica class.error
setosa         37          0         0  0.00000000
versicolor      0         31         1  0.03125000
virginica       0          1        35  0.02777778
```

For the random forests model, some key hyperparameters to look at is number of variables tried at each split (mtry), and number of trees (ntree). The number of variables tried at each split being set to 2 means that for each decision tree, only two out of the four variables are being randomly sampled as candidates. This hyperparameter matters because this is the way of random forests to build multiple different decision trees splits.

To evaluate this model on the training data, we can look at the Out-of-bag (OOB) estimate of error rate, as it was mentioned in regression. Since we know that the random forest algorithm uses bootstrapping sampling, and each bootstrap sample contain roughly 2/3 of uniquely data points, we can call that the remaining 1/3 are "out-of-bag." Therefore, the OOB estimate error will use each of these data points, gather all trees in the forest that did not use this specific point in their training, make a prediction with this forest, and then compare the predicted class to the true class. The formula for the OOB estimate error:

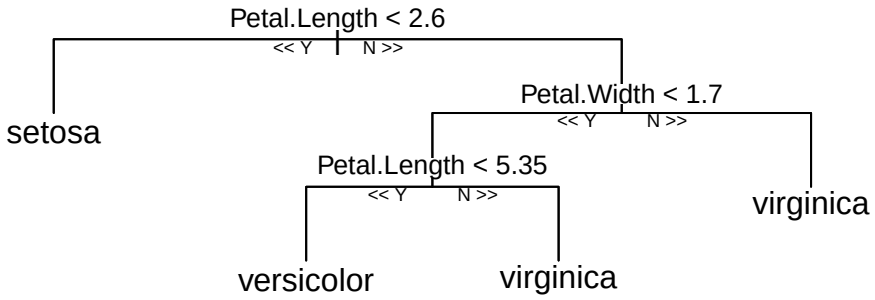
$\frac{1}{n}\sum_{i=1}^{n}\mathbf{I}\left(y_i \neq \hat{y}_i^{\text{OOB}}\right)$

$\hat{y}_i^{\text{OOB}}$ = The predicted class for the i-th data point. y_i = The true class for the i-th data point.

N = Total number of data points in the original training dataset. $\mathbf{I}$ = Indicator functions that will be equal to 1 if $(y_i \neq \hat{y}_i^{\text{OOB}})$ is true, and 0 otherwise.

The model achieved an Out-of-bag (OOB) error rate of 5.71 percent, meaning that 5.71 percent of the observation were misclassified when each point was predicted using only decision trees that did not include it during training.

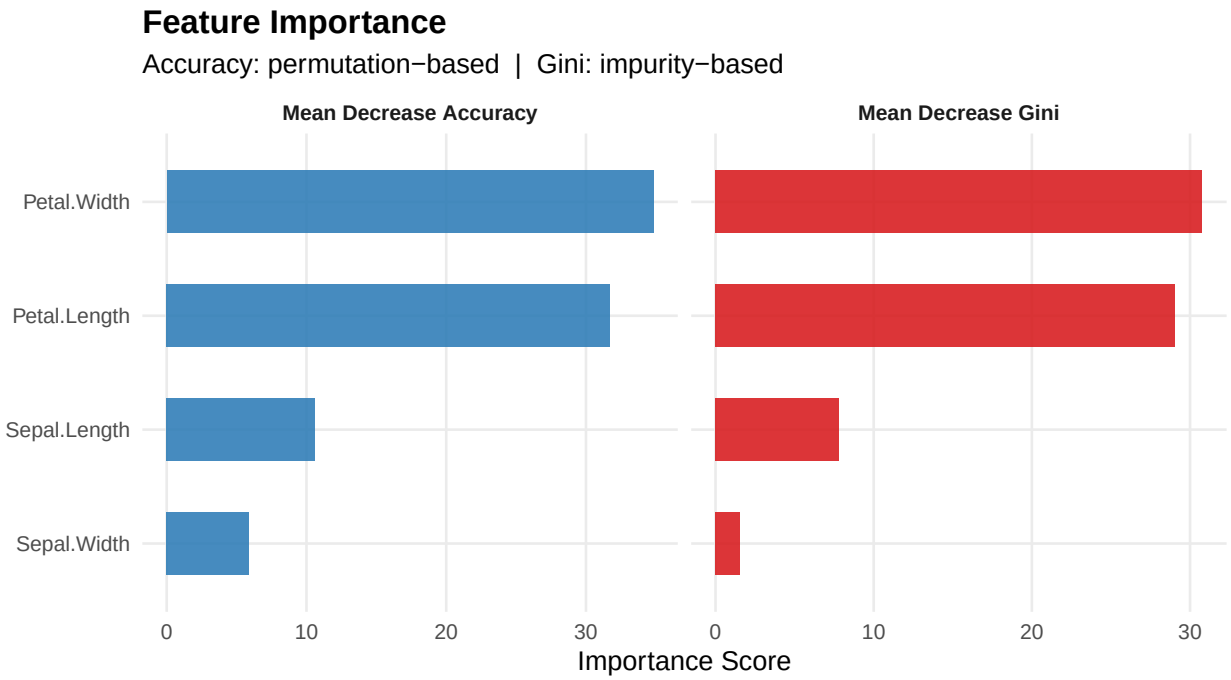
Singular Decision Tree



While visualizing a single decision tree does not explain the full behavior of the random forest, it provides useful intuition for how individual trees make predictions. Each tree is trained on a bootstrap sample of the data, and at every split, only a random subset of predictors is considered. From that subset, the algorithm selects the split that best separates the classes according to a criterion such as the Gini index previously mentioned.

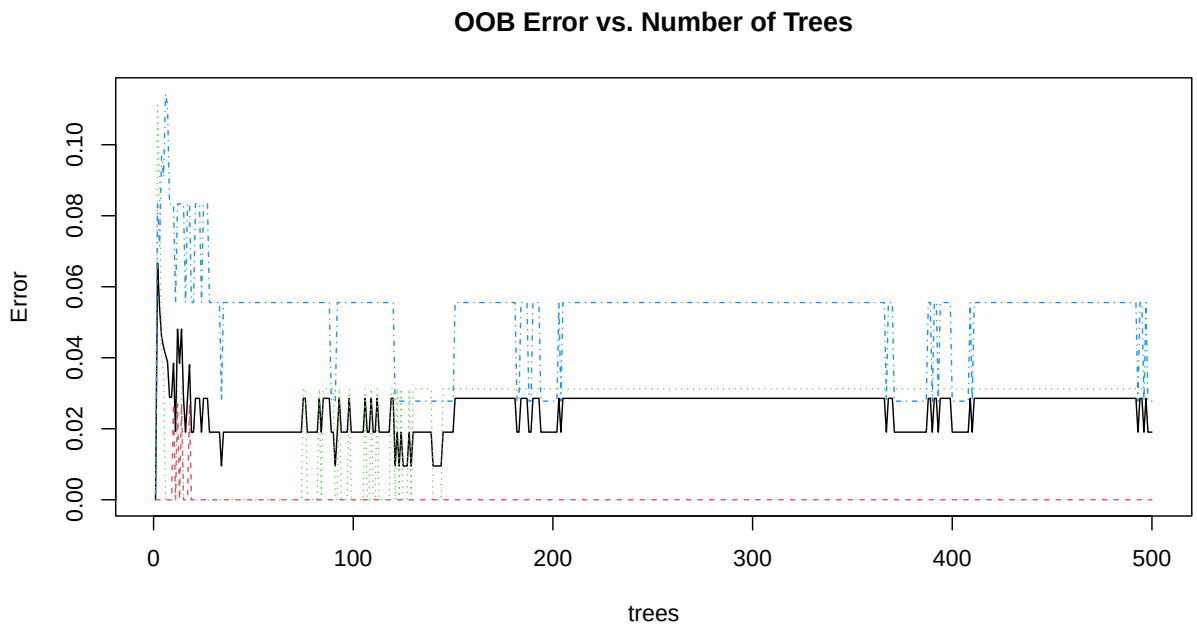
This means that no single feature is globally dominant in determining the structure of the tree, instead, different trees may use different features near the top depending on the random subset available at each split. The tree then recursively partitions the data by applying threshold rules on the selected features, creating decision paths that eventually lead to leaf nodes. Each leaf node assigns a predicted class, which in this dataset corresponds to one of the iris species.

Feature Influence



The feature importance plot shows a clear separation in how much each feature contributes to classifying iris species. The general idea here is that the Mean Decrease Gini is calculated by measuring how much that feature reduces uncertainty across all trees in the forest. Both metrics agree on the same overall structure: petal measurements dominate the model, while sepal measurements contribute much less. In particular, Petal Length and Petal Width are by far the most influential variables, meaning that when their values are used in splitting, model performance drops significantly.

Learning Curve — OOB Error vs. Trees



The out-of-bag (OOB) error curve highlights a pattern that is very typical for random forests: the model improves quickly at first and then levels off. Early in the process, when only a small number of trees have been built, the error rate fluctuates quite a bit. This instability is expected, since each prediction is based on a limited ensemble and is therefore more sensitive to the randomness in individual trees. As more trees are added, the error drops and the curve becomes noticeably smoother, reflecting the averaging effect that stabilizes the model's predictions.

After roughly 200 to 230 trees, the OOB error reaches a plateau and remains essentially constant. At this point, the model has captured most of the available signal in the data, and additional trees do not lead to meaningful improvements in performance. It is important to notice, the error does not increase as the number of trees grows, which is consistent with the idea that random forests do not overfit. In practice, this means that using a very large number of trees is often unnecessary, since a smaller forest can achieve the same predictive accuracy with less computational cost.

Model Evaluation

```
Confusion Matrix and Statistics

            Reference
Prediction   setosa versicolor virginica
  setosa         13          0         0
  versicolor      0         17         3
  virginica       0          1        11

Overall Statistics

               Accuracy : 0.9111
                 95% CI : (0.7878, 0.9752)
    No Information Rate : 0.4
    P-Value [Acc > NIR] : 9.959e-13

                  Kappa : 0.8645

 Mcnemar's Test P-Value : NA

Statistics by Class:

                     Class: setosa Class: versicolor Class: virginica
Sensitivity                 1.0000            0.9444           0.7857
Specificity                 1.0000            0.8889           0.9677
Pos Pred Value              1.0000            0.8500           0.9167
Neg Pred Value              1.0000            0.9600           0.9091
Prevalence                  0.2889            0.4000           0.3111
Detection Rate              0.2889            0.3778           0.2444
Detection Prevalence        0.2889            0.4444           0.2667
Balanced Accuracy           1.0000            0.9167           0.8767
```

The overall accuracy of 95.56% and a Kappa statistic of 0.93 both point to strong predictive performance. Looking at the class-level metrics, setosa again shows perfect scores, while versicolor and virginica have slightly lower sensitivity and precision. The only limitation is a pattern to confuse virginica with versicolor, which aligns with what was observed in the training results.

Conclusion

Random forests turn out to be a very solid model across both regression and classification problems, especially when the data has complex patterns that are hard to model directly. In the regression example, the model captured most of the variation in housing prices with strong agreement between out-of-bag and test error, which is a good indication that it is generalizing well rather than fitting noise. In the classification example, the model achieved high accuracy, with only a small amount of confusion between the more similar classes. Across both cases, the results show that random forests can deliver strong performance without requiring extensive tuning.

What stands out most is how the method balances flexibility and stability. By averaging many decision trees, it reduces the variability that makes single trees unreliable, while the randomness in the process helps prevent the model from relying too heavily on any one subset of features in the data. At the same time, there are clear limitations. In particular, random forests are not well suited for situations where interpretability is important, since the model's predictions come from an ensemble of many trees rather than a single, transparent structure. Although tools like variable importance provide some insight, they do not offer the same level of clarity as simpler models.

Overall, random forests are a dependable and practical tool, especially when the goal is accurate prediction and the underlying relationships in the data are not fully known. They may not be the best choice when clear model interpretation or extrapolation is required, but in many applied settings, they offer a strong balance of performance, robustness, and ease of use.